# BASE CHANGE OF A GYSIN MAP OF A LOCAL COMPLETE INTERSECTION MORPHISM IS THE TOP CHERN CLASS OF THE KERNEL OF THE CONORMAL COMPARISON

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

*Editorial note.* Canonical location of the counted proposition: Stacks tag 0FV9 (sources/stacks-chow/pinned-original-source.tex line 13738).

*Allowed standard prerequisites (editorial).*

- Finite local freeness of the kernel of the base changed conormal map. More on Algebra, Lemma labelled more-algebra-lemma-base-change-lci-bis, cited at proof lines 13763-13765. Provides the first assertion of the lemma.
- Factorization of a local complete intersection morphism, the normal cone presentation and base change of differentials and conormal sheaves. More on Morphisms, Lemma labelled more-morphisms-lemma-get-NL, cited at proof lines 13778-13782; Morphisms, Lemmas labelled morphisms-lemma-base-change-differentials and morphisms-lemma-conormal-functorial-flat, cited at proof lines 13786-13792. Identifies the two descriptions of C and lets the excess computation be applied.
- The excess formula for a regular immersion. Same chapter, Lemma labelled lemma-gysin-excess, cited at proof lines 13796-13798. Gives res(i^!) = c_top(C^vee) o (i')^! for the immersion part.
- Composition of the Gysin class of a local complete intersection morphism with the flat pullback. Same chapter, the identity f^! = i^! o h^* used at proof lines 13800-13805. Passes from the immersion to the full local complete intersection morphism.

*Source reference.* Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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*Date*: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0FV1: sources/stacks-chow/tags.txt, source lines 19796–19796. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13765 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0FV4: sources/stacks-chow/tags.txt, source lines 19799–19799. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13780 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 01V0: sources/stacks-chow/tags.txt, source lines 2429–2429. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13789 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0473: sources/stacks-chow/tags.txt, source lines 5494–5494. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13792 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0FV7: sources/stacks-chow/pinned-original-source.tex, source lines 12021–12021. Same-chapter pointer into the author environment lemma at source line 12021; invoked at proof line 13802 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

**Lemma 0.1.** *Let $(S, \delta)$ be as in Situation 02QL. Consider a cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

*of schemes locally of finite type over $S$. Assume*

(1) *both $f$ and $f'$ are local complete intersection morphisms, and*
(2) *the Gysin map exists for $f$*

*Then $\mathcal{C} = \mathrm{Ker}(H^{-1}((g')^* NL_{X/Y}) \to H^{-1}(NL_{X'/Y'}))$ is a finite locally free $\mathcal{O}_{X'}$-module, the Gysin map exists for $f'$, and we have*

$$res(f^!) = c_{top}(\mathcal{C}^\vee) \circ (f')^!$$

*in $A^*(X' \to Y')$.*

*Proof.* The fact that $\mathcal{C}$ is finite locally free follows immediately from More on Algebra, Lemma 0FV1. Choose a factorization $f = h \circ i$ with $h : P \to Y$ smooth and $i$ an immersion. Then we can factor $f' = h' \circ i'$ where $h' : P' \to Y'$ and $i' : X' \to P'$ the base changes. Picture

$$\begin{array}{ccccc} X' & \longrightarrow & P' & \longrightarrow & Y' \\ \downarrow & & \downarrow & & \downarrow \\ X & \xrightarrow{i} & P & \xrightarrow{h} & Y \end{array}$$

In particular, we see that the Gysin map exists for $f'$. By More on Morphisms, Lemmas 0FV4 we have

$$NL_{X/Y} = \left(\mathcal{C}_{X/P} \to i^*\Omega_{P/Y}\right)$$

where $\mathcal{C}_{X/P}$ is the conormal sheaf of the embedding $i$. Similarly for the primed version. We have $(g')^*i^*\Omega_{P/Y} = (i')^*\Omega_{P'/Y'}$ because $\Omega_{P/Y}$ pulls back to $\Omega_{P'/Y'}$ by Morphisms, Lemma 01V0. Also, recall that $(g')^*\mathcal{C}_{X/P} \to \mathcal{C}_{X'/P'}$ is surjective, see Morphisms, Lemma 0473. We deduce that the sheaf $\mathcal{C}$ is canonicallly isomorphic to the kernel of the map $(g')^*\mathcal{C}_{X/P} \to \mathcal{C}_{X'/P'}$ of finite locally free modules. Recall that $i^!$ is defined using $\mathcal{N} = \mathcal{C}_{Z/X}^\vee$ and similarly for $(i')^!$. Thus we have

$$res(i^!) = c_{top}(\mathcal{C}^\vee) \circ (i')^!$$

in $A^*(X' \to P')$ by an application of Lemma 0FV7. Since finally we have $f^! = i^! \circ h^*$, $(f')^! = (i')^! \circ (h')^*$, and $(h')^* = res(h^*)$ we conclude. □